Phang Hong Hao 2023080001

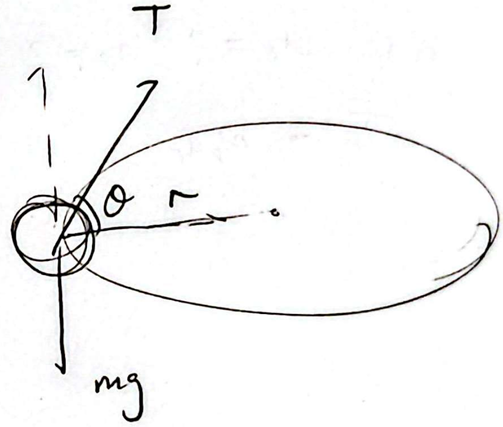
Q12.36

(a) $T_{BC} = ?$

Known $\rho = r = 0.93\,m$, $\theta = 60°$

$$\vec{a}_n = \frac{v^2}{r}\vec{e}_n$$

$$\begin{cases} \Sigma F_r = T\cos\theta = m a_n \\ \Sigma F_z = T\sin\theta - mg = 0 \end{cases}$$

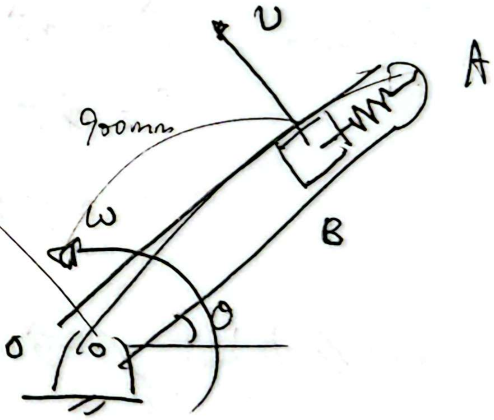


(a) $T = \dfrac{mg}{\sin\theta} = 80.4\,N$ #

(b) $\because \dfrac{mg}{\sin\theta}\cos\theta = m\left(\dfrac{v^2}{r}\right)$

$\therefore v = \sqrt{\dfrac{rg}{\tan\theta}} = 2.30\,m/s$ #

12.50 $\omega = \frac{v}{0.9} = 3.33\ \text{rad/s}$

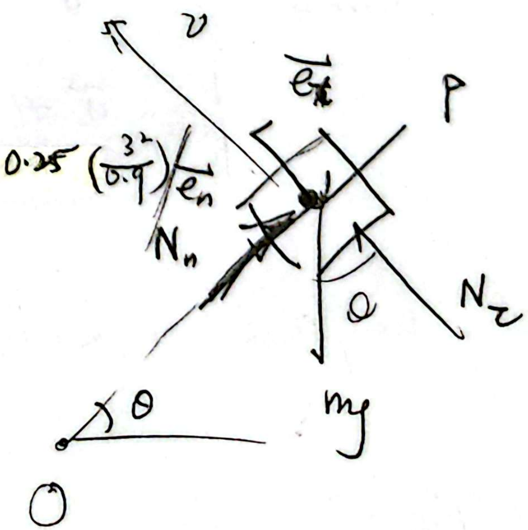


$\vec{a}_n = \frac{v^2}{r}\vec{e}_n$, $P = 1.5$, $mg = 0.25(9.8)$

$$\begin{cases} \Sigma F_n = -N_n + P + mg\sin\theta = ma_n & \text{①} \\ \Sigma F_t = N_t - mg\cos\theta = 0 & \text{②} \end{cases}$$

→ Contact with face of cavity closest to "O"

$N_n = 0$ (Critical Point)



① → $\Sigma F_n = -N_n + 1.5 + 0.25(9.8)\sin\theta = 0.25\left(\frac{3^2}{0.9}\right)\vec{e}_n$

Let $N_n = 0$,

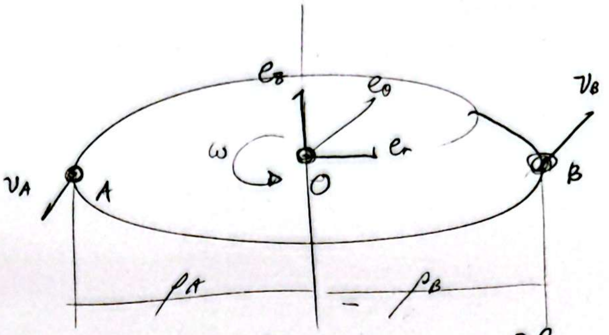
$\sin\theta = 0.408$

∴ $24.1^\circ \leq \theta \leq 155.9^\circ$

92 a) $a_r = ?$ $a_\theta = ?$

Q 12.92   a) $a_r = ?\ ,\ a_\theta = ?$

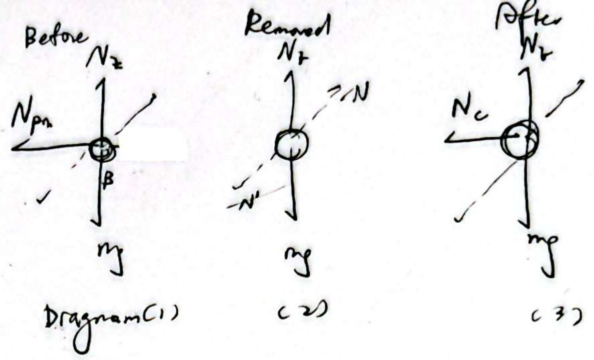
$$\omega = \frac{v_A}{\rho_A} = \frac{2.5}{0.25} = 10\ rad/s$$



Diagram(2), $a_r = 0 \quad (\because \Sigma F_r = 0)$

$a_\theta = 0$ # $\quad (\because \Sigma F_\theta = N - N' = 0)$ #

b) Relative to rod



$F_{inert} = m_B a_{relative}$

$\because F_{inert} = -N_{pm} = m \rho_B \omega^2$

$\cancel{m_B} a_{relative} = \cancel{m_B}(0.2)(10)^2$

$\therefore a_{relative} = 20 m/s^2$ #

c) $L = Const.$ (No exteral Torque exerted into system

$\therefore L_o = (\vec{\rho} \times m\vec{v})_A + (\vec{\rho} \times m\vec{v})_B$

$L' = (\vec{\rho} \times m\vec{v})_{A'} + (\vec{\rho} \times m\vec{v})_{B'}$

$L_o = L'$ $\qquad v_B = \rho_B \omega = (0.2)(10) = 2 m/s$

$\therefore (0.25)(0.2)(2.5) + (0.2)(0.4)(2) = (0.25)(0.2)(v_A') + (0.4)(0.4)($

$\because \omega = 10\ rad/s$ ~~(Const, $\Sigma = 0$)~~

$$\omega' = \frac{v_A'}{\rho_A} = \frac{v_B'}{\rho_B'} \quad \begin{cases} \rho_A = 0.2 \\ \rho_B' = 0.4 \end{cases}$$

$\therefore$

0.0285

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c) $L = L_o = L'$, $L = I\omega$

$I_o \omega_o = I'\omega'$

$(m_A r_A^2 + m_B r_B^2)\omega_o = (m_A r_A'^2 + m_B r_B'^2)\omega'$

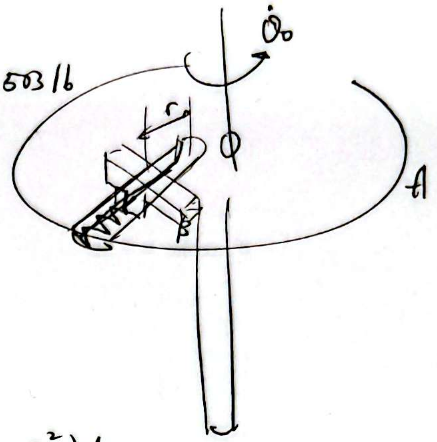
$[(0.2)(0.25)^2 + (0.4)(0.2)^2](10) = [(0.2)(0.25)^2 + (0.4)(0.4)^2]\omega'$

~~$\therefore \omega' = 1.92\ rad/s$~~ $\qquad \omega' = 3.725\ rad/s$

$$\because \omega' = \frac{v_A'}{\rho_A} = \frac{v_B}{\rho_B'} \quad \begin{cases} \rho_A = 0.25 \\ \rho_B = 0.4 \end{cases}$$

$\therefore v_A' = 6.93\ m/s$ #

12.131 $\vec{a}_B = \vec{a}_e + \vec{a}_r + \vec{a}_c$



(Known) $\dot{\theta}_0 = 12 \text{rad/s}$, $m = 8.05 \text{ oz} = 0.503 \text{ lb}$

$r = 0$, no exerted force,

$r = 15 \text{in}$, release,

$$\begin{cases} F = kr = m(\ddot{r} + r\dot{\theta}_0^2) \\ N = 2m\dot{\theta}_0 v_r \end{cases}$$

$$\int_{r_0}^{r} F dr = \int_{r_0}^{r} kr\, dr = \int_{r_0}^{r} m(\ddot{r} + r\dot{\theta}_0^2) dr$$

$$\frac{1}{2}k(r^2 - r_0^2) = \frac{1}{2}m(v_r^2 - 0^2) - \frac{1}{2}m\dot{\theta}_0^2(r^2 - r_0^2)$$

$$v_r = \sqrt{\frac{(k - m\dot{\theta}_0^2)(r_0^2 - r^2)}{m}} \quad , \quad r = \int_{t_0}^{t} v_r dt = \sin\left[\frac{\pi}{2} - \sqrt{\frac{(k - m\dot{\theta}_0^2)}{m}} \cdot t\right] r_0$$

a) $k = 2.25 \text{ lb/ft}$

$r = 15 \text{in}$, $N = 0 \text{ lb}$

b) $k = 3.25 \text{ lb/ft}$

$r = 10.5 \text{ in}$, $N = 2.69 \text{ lb}$